

SAI VARSHA SREEPERUMBUDUR

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[Publication IEEE](#)

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CORE COMPETENCIES

Programming Languages: Python(Pandas, Numpy,Scikit-Learn), R, SQL, Java, C, C++, Scala, Shell-Scripting, SAS, TypeScript

Data Analytics & Visualization: PowerBI (DAX, Data Modeling, Star Schema), Tableau, D3.js, Matplotlib, Seaborn, Plotly, Excel (Advanced), Streamlit

Databases & Data Storage:PostgreSQL, MySQL, SQL Server, MongoDB, DynamoDB, Neo4j, NoSQL Databases, Relational Databases, Data Lakes

Data Engineering & Integration: ETL / ELT Pipelines, Data Modeling, SSIS, SSRS, API Ingestion, REST APIs,Kafka (Event Streaming),Data Processing Pipelines, Batch Processing, Stream Processing, Data Validation, Data Quality Checks, Data Wrangling, Workflow Automation (n8n)

Machine Learning & Analytics: Regression, Classification, Clustering, KNN, SVM, Naive Bayes, Decision Trees, Ensemble Models (Random Forest, XGBoost, LightGBM, CatBoost), Time Series Forecasting (ARIMA, Prophet, GAM), Dimensionality Reduction (PCA, t-SNE), Hyperparameter Tuning, Model Evaluation (ROC-AUC, Precision, Recall, F1-Score), Prescriptive Analytics, TOPSIS

Big Data & Cloud Platforms: Apache Spark, Spark SQL, Apache Hadoop, AWS (S3, EC2, SageMaker), Google Cloud Platform, Azure, Microsoft Fabric (Data Lakes, Data Warehouses)

Deep Learning & NLP: CNN, RNN, LSTM, PyTorch, TensorFlow, Keras, NLTK, TF-IDF, Word Embeddings, BERT

Statistics & Mathematics: Probability, Descriptive Statistics, Inferential Statistics, Hypothesis Testing, Statistical Modeling for Analytics & Forecasting

AI Systems & Advanced Analytics: LLM Integration, Vector Databases, AI Workflow Orchestration, Intelligent Automation, Decision Support Systems, KNIME Analytics Platform

Software Development & Frameworks: ASP.NET Core, MVC Framework, Entity Framework Core, Dependency Injection, Middleware Configuration,React (Hooks, Context API), Web Applications **DevOps & Collaboration:** Git, GitHub, CI/CD Pipelines, Azure DevOps, Agile/Scrum, Jira, Technical Documentation

EXPERIENCE

AI Engineer-Data & Automation Systems - *Python, SQL,CRM, REST APIs, Power BI, Vector Database*

Hustad Companies Inc

(Jun 2025 - Present)

- Designed, built, and maintained automated data pipelines using Python, SQL, and REST APIs to ingest, transform, and load data from multiple enterprise systems.
- Orchestrated AI workflows integrating LLMs, vector databases, REST APIs, and workflow automation to deliver cohesive end-user experiences.
- Partnered across IT, product owners, and business stakeholders to contribute to solution paths for both technical and business problems.
- Participated in Agile ceremonies, including sprint planning, pointing, demos, and retrospectives.Supported production data issues by debugging pipeline failures and optimizing query and pipeline performance.
- Developed and maintained frontend components using React and TypeScript to visualize operational data and integrate seamlessly with backend REST APIs.
- Implemented API-driven state management and dynamic UI updates to support real-time data workflows and improve user interaction reliability.
- Contributed to the delivery pipeline using CI/CD best practices, leveraging Git for version control and deployment workflows.
- Implemented secure data handling workflows and access-controlled API integrations to support privacy-sensitive operational systems.

Graduate Data Science & Analytics Assistant :- *Python, SQL, R, Power BI, Tableau, Pandas, NumPy, Scikit-learn, Statistical Modeling, Machine Learning, Data Visualization*

University of Nebraska

(Feb 2024 – May 2025)

- Performed applied data analysis and machine learning on large structured datasets using Python, SQL, and R, including EDA, data cleaning, and feature engineering.
- Built and evaluated statistical and ML models (regression, classification, time-series forecasting) to support data-driven decision-making.
- Developed automated dashboards and reports in Power BI and Tableau for repeatable insight delivery.
- Communicated analytical findings to technical and non-technical stakeholders, translating results into actionable business insights.

Software Developer - *C#,Java, .NET, SQL Server,Cloud Platforms, MongoDB, REST APIs*

CGI

(Jun 2022 – Dec 2023)

- Developed backend data loads and integrations using SQL Server (MSSQL) in a .NET enterprise environment.

- Designed and presented in-depth instructional content on Software System Design, emphasizing architectural patterns, design principles, and system scalability; boosted business proficiency, reflected in a 15% rise in production.
- Automated ETL processes and data workflows to improve data reliability and system performance.
- Worked in Agile teams to deliver production-ready solutions with a focus on scalability and data quality.

PROJECTS

Child Care Staffing Forecasting& Scheduling

Jan 2025 - May 2025

Python, SQL, PyTorch, Scikit-learn, GAM, Random Forest, AWS (S3, SageMaker), Power BI, Streamlit

- Built an end-to-end analytics pipeline transforming raw attendance events into structured time-series datasets for workforce demand forecasting.
- Engineered features and evaluated multiple ML approaches (GAM, Random Forest, PyTorch-based models) using back-testing to compare forecast accuracy and stability.
- Automated reporting through Power BI dashboards and developed an interactive Streamlit application (`Dashboard_Tools.py`) for forecast visualization and decision support.
- Implemented a reproducible modeling workflow with consistent data splits and evaluation metrics to ensure reliable and comparable forecasting results over time.

Sales Analytics& Business Intelligence Dashboards

Sep 2024 - Dec 2024

Power BI, SQL, Data Modeling, Advanced Query, DAX, ETL

- Designed analytics-ready relational datasets using SQL transformations and star-schema data modeling.
- Developed Power BI semantic models with optimized relationships and DAX measures for revenue, growth, and performance tracking.
- Automated data refresh and reporting workflows to enable repeatable, self-service analytics.
- Delivered interactive dashboards supporting business performance monitoring and executive decision-making.
- Optimized dashboard performance and query efficiency to support scalable reporting across multiple business dimensions.

Customer Churn Prediction & Behavioral Analytics

Jun 2024 - Aug 2024

Python, SQL, Scikit-learn, Feature Engineering, Classification Models

- Performed exploratory data analysis and feature engineering on customer behavioral datasets to identify churn-driving patterns.
- Built and evaluated classification models (Logistic Regression, Random Forest) using ROC-AUC, Precision, Recall, and F1-score.
- Translated model outputs into actionable insights to support customer retention and engagement strategies.
- Assessed feature importance and model behavior to improve interpretability and support data-driven retention decisions.

Greenhouse Gas Emissions Trend & Forecast Analysis

Feb 2024 - Aug 2024

Python, SQL, Time Series Analysis, Data Visualization (Plotly, Power BI)

- Integrated multi-year U.S. emissions datasets and engineered structured time-series data for trend analysis.
- Applied time-series modeling to analyze historical patterns and forecast emissions trends across regions.
- Delivered interactive visualizations and dashboards to support data-driven environmental insights and reporting.
- Validated forecasting outputs against historical benchmarks to ensure trend consistency and analytical reliability.

EDUCATION

University of Nebraska at Omaha, Nebraska

Jan 2024 - Dec 2025

Master of Science in Data Science (IT Concentration)

- **Coursework:** Artificial Intelligence & Machine Learning, Data Visualization, Cloud Computing, Programming & Problem Solving, Software Development,

ACHIEVEMENTS & CERTIFICATIONS

- **2nd Place – FNBO Datathon (2024):** Analyzed FNBO reward programs (Getaway, Evergreen) to identify patterns in customer engagement, spending behavior, and redemption trends; built ML-based customer segmentation models and delivered interactive Power BI dashboards with data-driven recommendations.
- **UNO Data Mining Competitions – Spring 2025:**
 - **1st Place, Contest#2 (15 teams):** Built and tuned predictive ensemble ML models, achieving **97% accuracy** on real-world datasets.
 - **2nd Place, Contest#1 (6 teams):** Applied advanced data mining and model tuning techniques to generate actionable insights.
- **Microsoft Certified:** Azure Fundamentals (AZ-900), Azure AI Fundamentals (AI-900).
- **Power BI Certification:** Data modeling, DAX measures, and dashboard development.
- **Databricks Generative AI Fundamentals:** Foundational training in generative AI concepts and applications.